

Homework 1 Group I

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Question 1

Let $s_i = 1 - 2y_i$. For $\lambda > 0$, the objective function is

$$f_\lambda(\theta) = \underbrace{\sum_{i=1}^m \phi(s_i \langle \tilde{x}_i, \theta \rangle)}_{f(\theta)} + \frac{\lambda}{2} \|\theta\|^2, \quad \theta \in E.$$

We first show that f is convex. The smoothed hinge loss ϕ satisfies

$$\phi'(z) = \begin{cases} 0, & z \leq -1, \\ 1 + z, & -1 < z < 0, \\ 1, & z \geq 0. \end{cases}$$

The values and piecewise derivatives agree at $z = -1$ and $z = 0$. Thus ϕ is continuously differentiable. Its derivative is nondecreasing, so for any $a, b \in \mathbb{R}$,

$$\phi(b) - \phi(a) - \phi'(a)(b - a) = \int_a^b (\phi'(u) - \phi'(a)) du \geq 0.$$

(The inequality holds for either ordering of a and b .) By Theorem 4.21 of the lecture notes, ϕ is convex.

For any $\theta, \eta \in E$ and $t \in [0, 1]$, linearity of the inner product and convexity of ϕ give

$$\begin{aligned} \phi(s_i \langle \tilde{x}_i, (1-t)\theta + t\eta \rangle) &= \phi((1-t)s_i \langle \tilde{x}_i, \theta \rangle + ts_i \langle \tilde{x}_i, \eta \rangle) \\ &\leq (1-t)\phi(s_i \langle \tilde{x}_i, \theta \rangle) + t\phi(s_i \langle \tilde{x}_i, \eta \rangle). \end{aligned}$$

Summing over i yields $f((1-t)\theta + t\eta) \leq (1-t)f(\theta) + tf(\eta)$; hence f is convex.

Now

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex. Since $\lambda > 0$, Definition 4.5 of the lecture notes therefore implies that f_λ is λ -strongly convex. It is real valued on the whole Euclidean space E , so Theorem 4.28 applies: f_λ has a unique global minimizer θ^* .

Moreover, f_λ is convex and differentiable on E . Each loss term is a composition of the continuously differentiable function ϕ with a linear map, and the quadratic term is differentiable. By Corollary 4.23 of the lecture notes, a point is a global minimizer if and only if its gradient vanishes. So, what we can say about the critical points is that

$$\{\theta \in E : \nabla f_\lambda(\theta) = 0\} = \{\theta^*\}.$$

Question 2

We now set $\lambda = 0.005$. Write $g_i(\theta) = s_i \langle \tilde{x}_i, \theta \rangle$. Each $g_i : E \rightarrow \mathbb{R}$ is linear, hence differentiable, and ϕ is continuously differentiable as shown in Question 1. Thus $f = \sum_{i=1}^m \phi \circ g_i$ is differentiable. By additivity of the differential and the chain rule, for any $h \in E$,

$$Df(\theta)[h] = \sum_{i=1}^m \phi'(g_i(\theta)) Dg_i(\theta)[h].$$

Using the limit expression for the differential of g_i , we obtain

$$Dg_i(\theta)[h] = \lim_{t \rightarrow 0} \frac{g_i(\theta + th) - g_i(\theta)}{t} = \lim_{t \rightarrow 0} \frac{ts_i \langle \tilde{x}_i, h \rangle}{t} = s_i \langle \tilde{x}_i, h \rangle.$$

Substituting and using bilinearity of the inner product gives

$$\begin{aligned} Df(\theta)[h] &= \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \langle \tilde{x}_i, h \rangle \\ &= \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i, h \right\rangle. \end{aligned}$$

The quadratic term $r(\theta) = \frac{\lambda}{2} \|\theta\|^2$ is also differentiable. Indeed,

$$r(\theta + h) - r(\theta) - \lambda \langle \theta, h \rangle = \frac{\lambda}{2} \|h\|^2 = o(\|h\|),$$

so $Dr(\theta)[h] = \lambda \langle \theta, h \rangle$. Consequently, $f_\lambda = f + r$ is differentiable and

$$Df_\lambda(\theta)[h] = \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta, h \right\rangle.$$

By Definition 2.18 of the lecture notes, $Df_\lambda(\theta)[h] = \langle \nabla f_\lambda(\theta), h \rangle$. Since this equality holds for every $h \in E$, we conclude that

$$\nabla f_\lambda(\theta) = \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta.$$

Numerical agreement. We evaluated the loop and vectorized implementations on the same training data and parameter vector (random seed 329), with $\lambda = 0.005$. Using NumPy's `assert_allclose`, we compared the objective values and each gradient component with absolute and relative tolerances of 10^{-10} : a pair a, b , with b the vectorized result, passes when $|a - b| \leq 10^{-10} + 10^{-10}|b|$. Both checks passed. The absolute objective difference was 5.093×10^{-11} , and the largest absolute gradient-component difference was 1.569×10^{-10} , consistent with floating-point roundoff from different evaluation orders.

Question 4

Given that ∇f_λ is L -Lipschitz continuous with constant $L = \sigma_{\max}(X)^2 + \lambda$ (where $\sigma_{\max}(X)$ is computed numerically), we select the constant step-size $\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda}$ as a consequence of Theorems 3.3 and 3.4. . Given that ∇f_λ is L -Lipschitz continuous with constant $L = \sigma_{\max}(X)^2 + \lambda$ (where $\sigma_{\max}(X)$ is computed numerically), we select the constant step-size $\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda}$ as a consequence of Theorems 3.3 and 3.4. .

TODO : USE THEOREM 4.31 INSTEAD AS IT HAS A CRITERION FOR CONVERGENCE LINEARLY!

TODO : CHANGE THE README EXPLANATION !!

Question 6

Consider the fixed-step gradient descent iteration

$$\theta_{k+1} = \theta_k - \alpha \nabla f_\lambda(\theta_k), \quad k \geq 0.$$

From Question 1, f_λ is λ -strongly convex since

$$f_\lambda(\theta) - \frac{\lambda}{2}\|\theta\|^2 = f(\theta)$$

is convex. Moreover, as stated in the homework,

$$\nabla f_\lambda$$

is Lipschitz continuous with constant

$$L = \sigma_{\max}(X)^2 + \lambda.$$

Hence the assumptions of Theorem 4.31 are satisfied with

$$\mu = \lambda, \quad L = \sigma_{\max}(X)^2 + \lambda.$$

Therefore f_λ has a unique minimizer θ^* , and for the step size

$$\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda},$$

Theorem 4.31 gives

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq \left(1 - \frac{\lambda}{L}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

Thus the objective values converge globally and at least linearly to the optimum.

More generally, for every

$$0 < \alpha \leq \frac{1}{L},$$

the same theorem may be applied with Lipschitz constant $1/\alpha$, yielding

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq (1 - \alpha\lambda)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

For the value $\lambda = 0.005$, the admissible step sizes are therefore

$$0 < \alpha \leq \frac{1}{\sigma_{\max}(X)^2 + 0.005}.$$

Application to our implementation

Our implementation uses exactly

$$\alpha = \frac{1}{L}, \quad L = \sigma_{\max}(X)^2 + \lambda,$$

so Theorem 4.31 applies directly.

Writing

$$\kappa = \frac{L}{\lambda},$$

we obtain

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq \left(1 - \frac{1}{\kappa}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

Corollary 4.32 further gives

$$\|\theta_k - \theta^*\| \leq \sqrt{\frac{2}{\lambda} (f_\lambda(\theta_0) - f_\lambda(\theta^*))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k$$

and

$$\|\nabla f_\lambda(\theta_k)\| \leq \sqrt{2L (f_\lambda(\theta_0) - f_\lambda(\theta^*))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k.$$

Hence both the iterates and the gradient norm converge at least linearly.

Comparison with the finite run

This prediction is consistent with our numerical experiment. The plot shows $\|\nabla f_\lambda(\theta_k)\|$ decreasing rapidly at first and then more slowly, until the stopping tolerance is reached after approximately 2.7×10^4 iterations.

Corollary 4.32 only provides the geometric upper bound

$$\|\nabla f_\lambda(\theta_k)\| \leq Cq^k, \quad q = \sqrt{1 - \frac{\lambda}{L}} < 1,$$

rather than requiring the gradient norm to decrease by the same factor at every iteration. Thus the changes of slope and relatively flat regions in the plot are compatible with the theory.

Since the numerical run stops once the prescribed tolerance is reached, it does not show convergence all the way to zero, but its behavior is consistent with the asymptotic prediction

$$\|\nabla f_\lambda(\theta_k)\| \rightarrow 0.$$

Question 7

The classification error is approximately 0.134% on the training set and 0.189% on the test set. The test error is slightly higher than the training error, as expected, but remains very small, indicating that the classifier generalizes well to unseen samples.

Q8. Mathematical insight

The regularization term

$$\frac{\lambda}{2} \|w\|^2$$

makes f_λ λ -strongly convex. Therefore, f_λ has a unique minimizer w^* , and gradient descent converges to w^* from any starting point. In particular, with the step size

$$\alpha = \frac{1}{L},$$

where L is the Lipschitz constant of ∇f_λ , gradient descent converges at least linearly.

Computational insight

Vectorizing the computations significantly improves the efficiency of the implementation. Instead of evaluating the objective and gradient with explicit Python loops, we use matrix operations such as

$$X^\top(X\theta - y).$$

These operations are executed by optimized numerical linear algebra routines and are therefore much faster.

In our experiments, the vectorized implementation was about 8 times faster for the objective function and about 10 times faster for the gradient, while producing numerically equivalent results. Since gradient descent requires many iterations, this speed-up has a substantial impact on the total runtime.